

# XIUCHUNLAI

## SUTURE LINE

Stitching the fractured city and its social life back together

— Life is tattered; AI stitches it up —

Centennial Beijing–Zhangjiakou AI Innovation Belt Open Urban Design Call • Submission

A century ago, the “ren” (人) character conquered the mountain; a century later, the “ren” character stitches people together.

### Needle-spot spacing

≈ 2-min walk; design hypothesis to be validated in pilots, not a proven “optimum”

### 15-min encounter coverage

Illustrative 750 m Euclidean-buffer coverage of meeting nodes, not a network 15-min accessibility result

### Public activity space / person

Green + plazas 78.7 ha → 70,000 directly served residents (assumed)

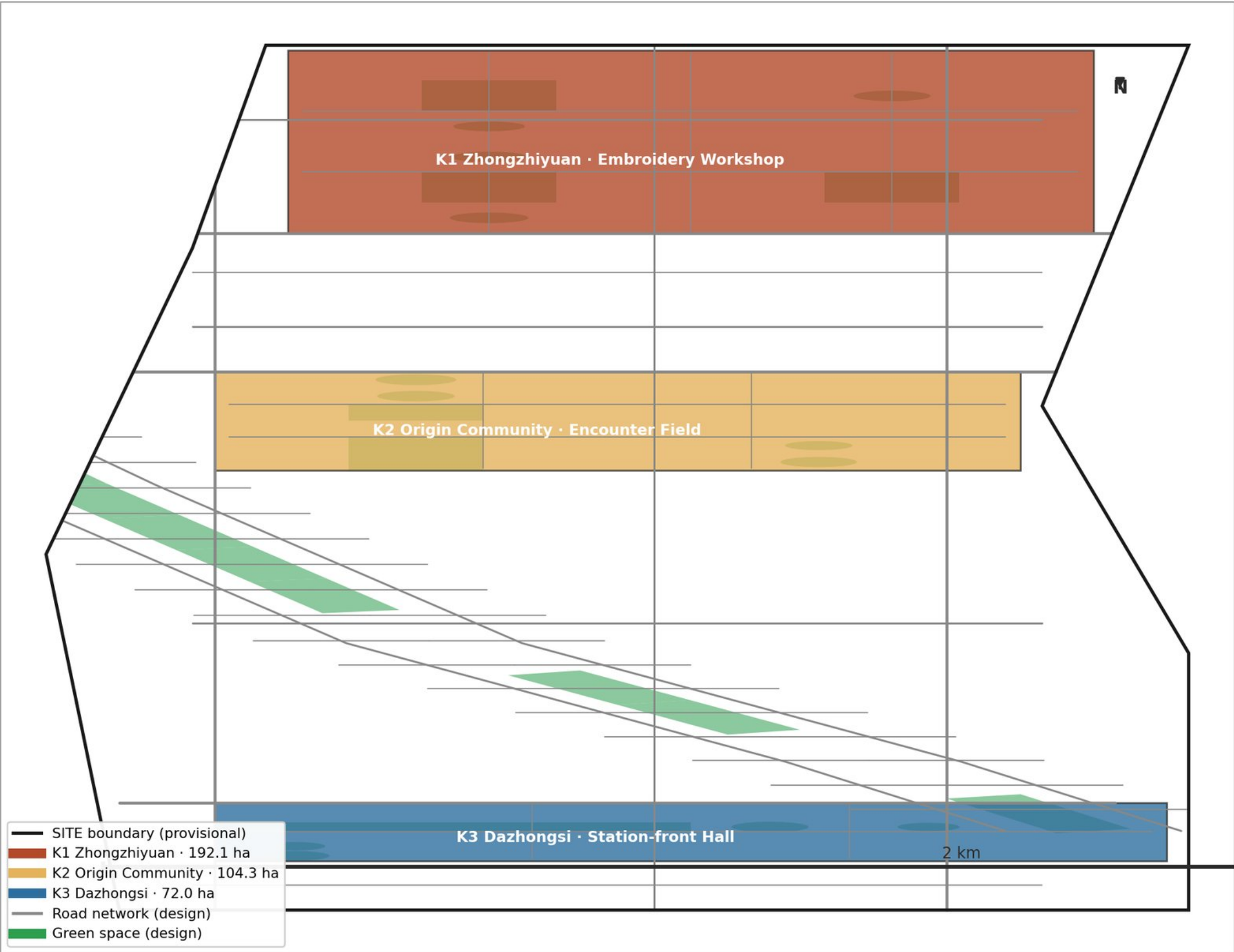
Three highlights • Embroidered Chat Furniture (200 LLM terminals) | Token-generating Embroider Bikes | Embroidered Phrase Garden (Transformer landmark)

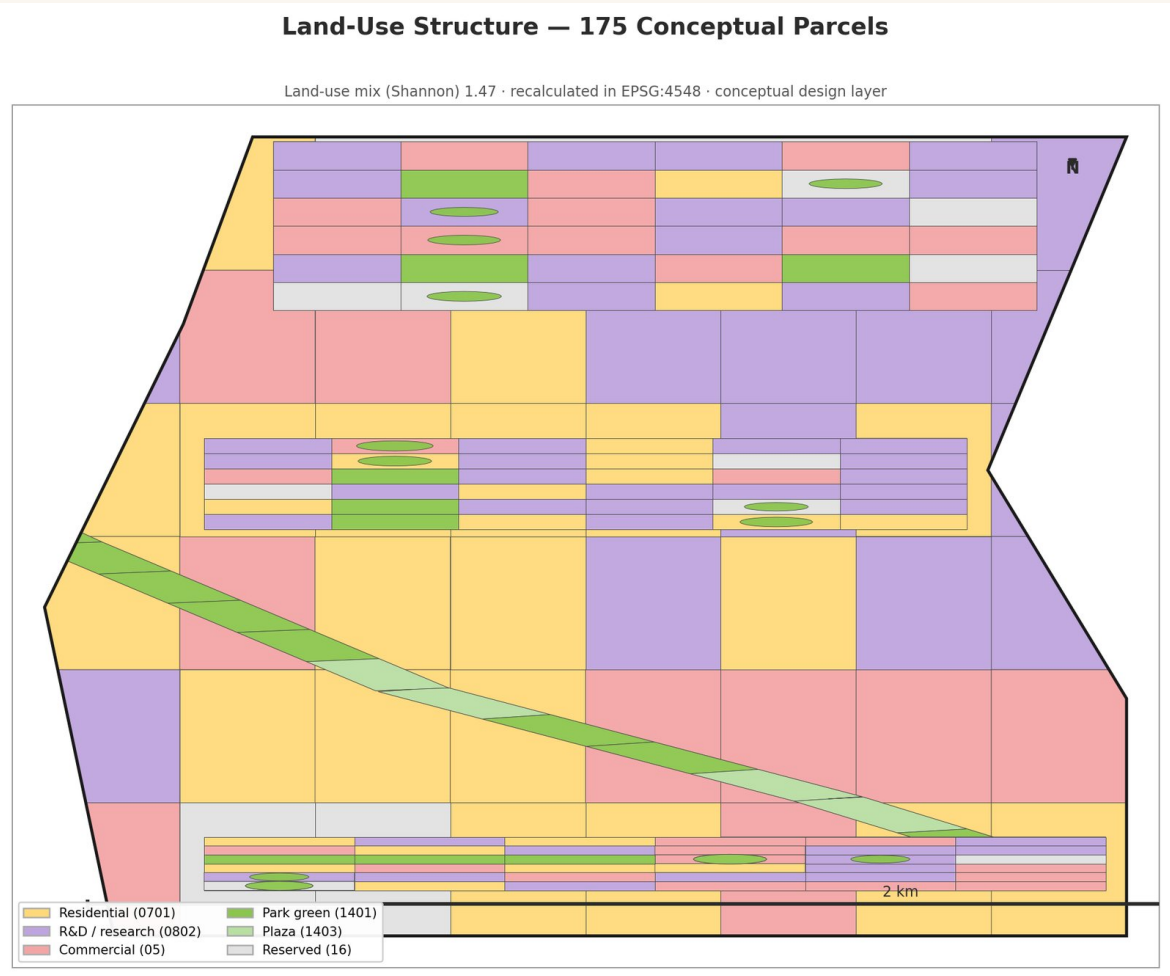
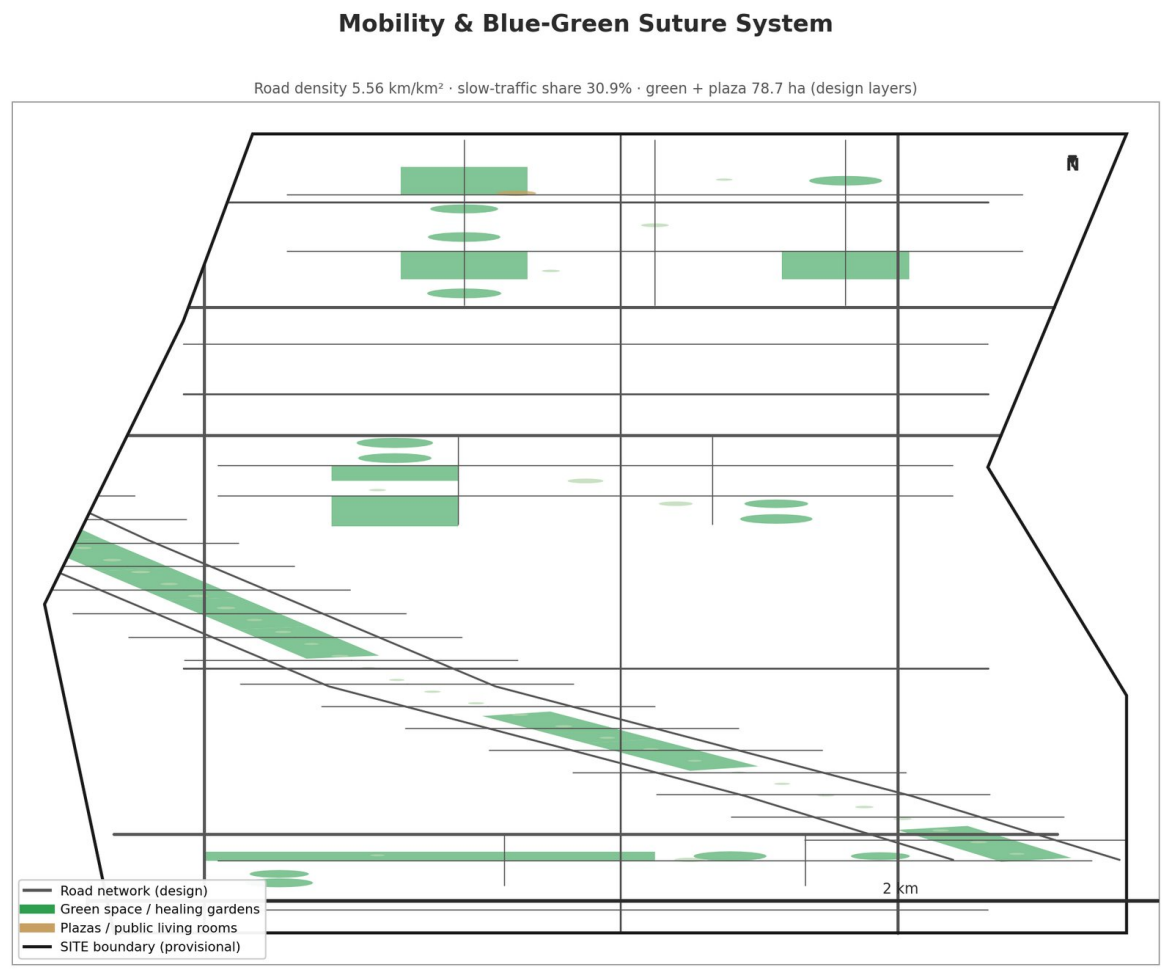
Three key areas • K1 Embroidery Workshop | K2 Encounter Field | K3 Station-front Hall

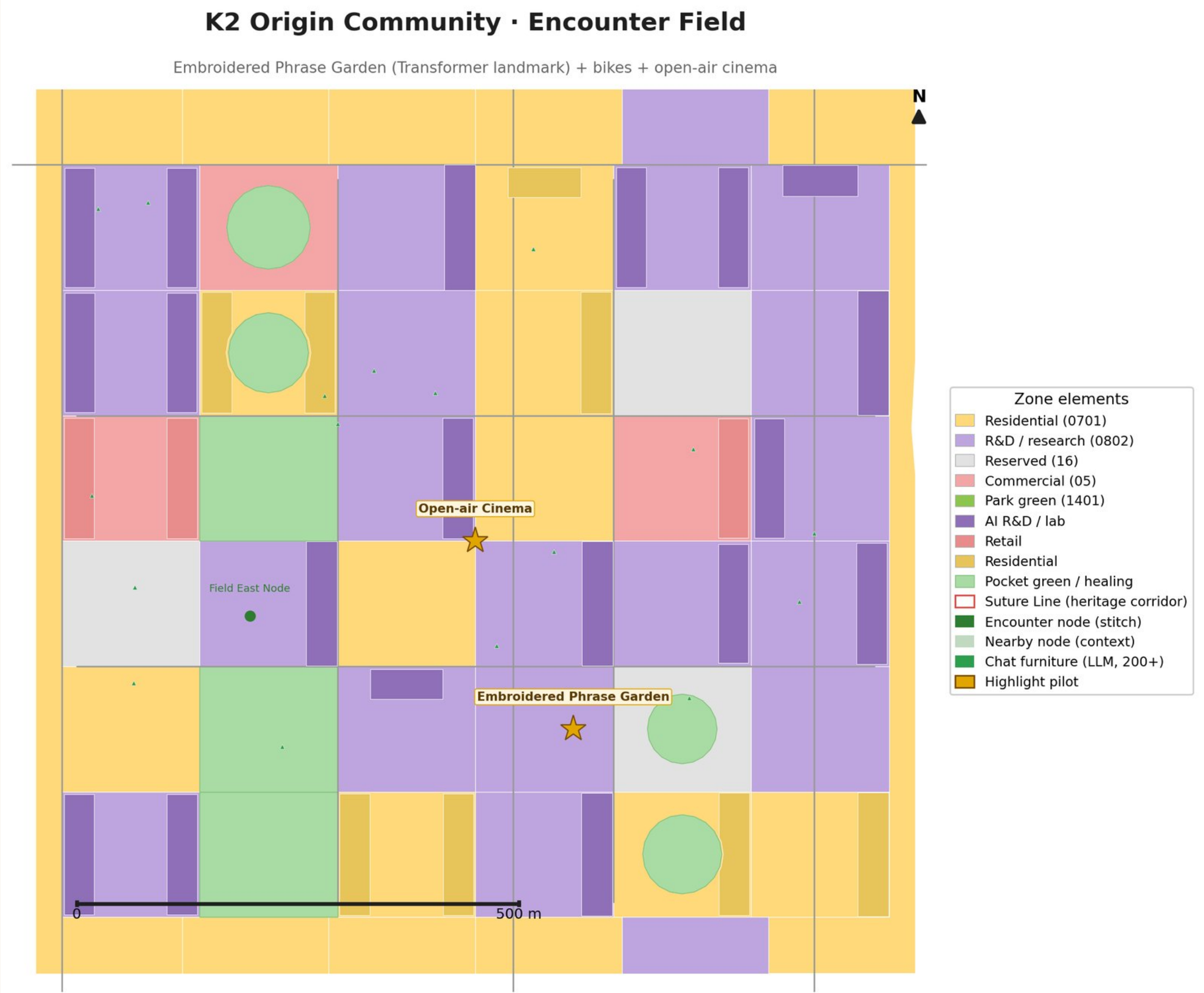
Four stitch layers • Encounter → Shared activity → Healing → Warmth (throughout)

# Suture Line — Overall Structure & Three-Level Scope

Provisional boundary PROV-SITE-001 (EPSG:4548 recomputed) · official red line pending







Highlights are scenario illustrations (see scenario cards), not engineering drawings; encounter-node spacing  $\approx 150$  m (about a 2-min walk).

Extent is the provisional boundary (official\_boundary=false); recompute after red lines; no commercial basemap used.

## Indicator Evidence Chain — 43 Indicators, Item-by-Item Source Files

Every indicator in metrics.json registers its method, source file and formula; recalculated values, declared values and targets awaiting verification are distinguished.

### Geometry recalculated (EPSG:4548)

|     |                              |             |                       |
|-----|------------------------------|-------------|-----------------------|
| A1  | SITE area                    | 1,141.5 ha  | site_boundary.geojson |
| B6  | Total road length            | 63.42 km    | roads.geojson         |
| B7  | Road density                 | 5.56 km/km² | roads.geojson         |
| B12 | land-use mix (Shannon)       | 1.47        | land_use.geojson      |
| E30 | Belt-wide green ratio        | 6.9%        | green_space.geojson   |
| F37 | Building coverage            | 7.2%        | buildings.geojson     |
| F39 | Building volume in key areas | 92.8%       | buildings.geojson     |

### Declared / target values (to verify)

|     |                                  |           |                      |
|-----|----------------------------------|-----------|----------------------|
| C14 | Stitch spacing (concept target)  | 1.5cm     | assumption - pilot   |
| C15 | Encounter coverage (750m buffer) | 71.7%     | algorithmic, Euclid. |
| C16 | Public space per person          | 11.2 m²/p | area ÷ 70k           |
| D26 | AI scenario cards                | 12        | constraints.geojson  |
| D28 | Industry test scenarios          | 3         | constraints.geojson  |
| A5  | Chat furniture density           | 17.5/km²  | constraints.geojson  |

Population is an assumption (K1≈20k / K2≈30k / K3≈20k, total 70k); all area and density values are recalculated under the design-layer caliber and do not constitute statutory planning indicators; official red line release triggers full recalculation.

Metrics & headline numbers (43 items; recomputed / declared / to-be-verified distinguished)

|                     |                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| A Overview          | Area 1141.3 ha   Spine 4.03 km   Key areas 3   Needles 31   Chat furniture 200                                |
| B Spatial structure | Road density 5.56 km/km²   Slow-traffic 30.9%   Needle lanes 29   Mixed-use 1.32                              |
| C Social capital    | ★ Needle spacing 150 m (concept)   ★ Encounter coverage 71.7% (750 m buffer, illustrative)   ★ 11.2 m²/person |
| D AI facilities     | Chat furniture density 17.5/km²   AI scenario cards 12   Pilot landmarks 5                                    |
| E Green & blue      | Green+plaza 78.7 ha   Healing green 29 sites   Key-area green ratio 10.0/11.4/16.4%                           |
| F Buildings         | 97 conceptual blocks   Coverage 7.2%   Avg depth 25.8 m   Key-area building share 93.8%                       |
| G Honest compliance | Manual fallback 100%   Emergencies bypass AI   Power output symbolic   Provisional boundary to be recomputed  |

★ Headline numbers (for oral defense) • Population assumed (K1≈20k/K2≈30k/K3≈20k, 70k total); recompute after official data release

16 geometric validations all PASS (2026-08-30): legal enumeration / 97 blocks non-overlapping / roads unique & in-bounds / needle spacing 150 m / 200 chat furniture compliant / key-area green ratio ≥8%